3.28 Regular Local Rings

The results of the previous section allow a more direct characterization of the dimension of a Noetherian local ring, which naturally leads to the notion of regular local ring.

Lemma 3.28.1 ($\mathfrak{m}$-primary)
Let $(A, \mathfrak{m})$ be a local ring and $\mathfrak{a}$ a proper ideal. Then the following are equivalent

a) $\sqrt{\mathfrak{a}} = \mathfrak{m}$
b) $\mathfrak{m}$ *is a minimal prime of* $\mathfrak{a}$

Furthermore $\mathfrak{a}$ is primary. In this case we say $\mathfrak{a}$ is $\mathfrak{m}$-primary. A sufficient condition is that for some n

$$\mathfrak{m}^n \subset \mathfrak{a} \subset \mathfrak{m}$$

and this is necessary when $\mathfrak{m}$ is finitely generated.

Proof. a) $\implies$ b) Suppose $\mathfrak{a} \subseteq \mathfrak{p}$ then $\mathfrak{m} = \sqrt{\mathfrak{a}} \subseteq \mathfrak{p}$ by (3.3.45). In particular $\mathfrak{m}$ is a minimal prime.

b) $\implies$ a) Let $\mathfrak{p}$ be a prime ideal containing $\mathfrak{a}$. By (3.3.36) $\mathfrak{p} \subseteq \mathfrak{m}$ and by assumption $\mathfrak{p} = \mathfrak{m}$. The result then follows from (3.3.45).

Suppose $xy \in \mathfrak{a}$ and $y \notin \sqrt{\mathfrak{a}}$ then by (...) $y \in A^\star$ whence $x \in \mathfrak{a}$. This shows that $\mathfrak{a}$ is primary.

The condition $\mathfrak{m}^n \subset \mathfrak{a}$ is clearly sufficient, and necessary in the finitely generated case by (3.3.47). ∎

An ideal satisfying one of the equivalent conditions above is called **$\mathfrak{m}$-primary**, though some authors require the stronger condition.

Proposition 3.28.2 (Criteria for Dimension of Local Ring)
Let $(A, \mathfrak{m})$ be a Noetherian local ring satisfying the generalized hauptidealsatz. Then we have the following criteria for dimension

$$\dim A = \operatorname{ht}(\mathfrak{m}) = \min\left\{ n \;\middle|\; \sqrt{(x_1, \ldots, x_n)} = \mathfrak{m} \right\} \leq \mu(\mathfrak{m}) = \dim_{k(\mathfrak{m})} \mathfrak{m}/\mathfrak{m}^2$$

where $\mu(\mathfrak{m})$ is the least number of generators of $\mathfrak{m}$ and $k(\mathfrak{m}) = A/\mathfrak{m}$.

Proof. The first equality follows because every maximal chain must terminate at a maximal ideal by (3.3.36). The second from the characterization of height for hauptidealsatz rings (3.27.12) and the equivalent definitions of $\mathfrak{m}$-primary (3.28.1). The inequality follows because $\mathfrak{m}$ is itself $\mathfrak{m}$-primary. The final equality follows because a minimal generating set lifts to a basis of $\mathfrak{m}/\mathfrak{m}^2$ (3.21.7). ∎

Definition 3.28.3
*Let $(A, \mathfrak{m})$ be a Noetherian local ring. We say A is **regular** if*

$$\dim A = \mu(\mathfrak{m}) = \dim_{k(\mathfrak{m})} \mathfrak{m}/\mathfrak{m}^2$$